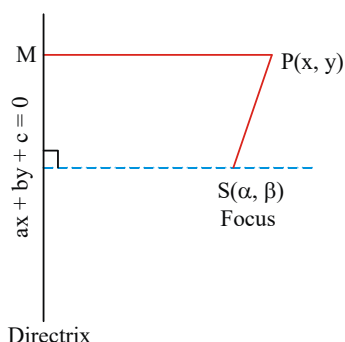


PARABOLA



SYNOPSIS

- **Definition of a Conic :** The locus of a point which moves in a plane such that the ratio of its distance from a fixed point to its perpendicular distance from a fixed straight line is always a constant, is called a conic section or a conic.



Let S be a fixed point and L be a fixed line, P be any point in the plane. Let M be the projection of P

on L. Then the locus of P such that $\frac{SP}{PM} = e$

constant, is called a conic. The fixed point S is called the **focus**, the fixed line L is called the **directrix** and the constant ratio is denoted by e, is called the **eccentricity** of the conic.

If $e = 1$, the conic is a parabola

If $0 < e < 1$, the conic is an ellipse

If $e > 1$, the conic is a hyperbola

If $e = 0$, the conic is a circle

If $e \rightarrow \infty$, the conic is a pair of straight lines.

- **General Equation of a Conic**

Equation of a conic with (x_1, y_1) focus $lx + my + n = 0$ directrix and eccentricity e is

$$(l^2 + m^2)[(x - x_1)^2 + (y - y_1)^2] = e^2 (lx + my + n)^2.$$

- The general equation of a conic is of the form $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$.

- **Recognition of Conics :** The equation of a conic is represented by the following general equation of second degree :

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \dots 1$$

Case 1 : When

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

Equation (1) represents a pair of straight lines.

Condition

Conic

- $\Delta = 0$ & $h^2 = ab$ a pair of parallel lines
- $\Delta = 0$ & $h^2 > ab$ a pair of intersecting lines
- $\Delta = 0$ & $h^2 < ab$ a pair of imaginary lines intersecting at a real point

Case 2: If $\Delta \neq 0$

Condition

Conic

- $\Delta \neq 0, a = b \neq 0, h = 0$ a circle
- $\Delta \neq 0, h^2 = ab$ a parabola
- $\Delta \neq 0, h^2 < ab$ an ellipse (or an empty set)
- $\Delta \neq 0, h^2 > ab$ a hyperbola
- $\Delta \neq 0, h^2 > ab, a + b = 0$ a rectangular hyperbola

Definitions :

- Principal Axis (Axis):** The straight line passing through the focus and perpendicular to the directrix is called the axis of the conic section.
- Vertex (Vertices):** The point(s) of intersection of the conic section and the axis is (are) called the vertex (vertices) of the conic.
- Chord :** The line segment joining any two points on the conic is called a chord.
- Focal chord :** Any chord passing through the focus is called a focal chord of the conic section.
- Double ordinate :** A chord passing through a point P on the conic and perpendicular to the axis of the conic is called the double ordinate of the point P.
- Latus rectum :** The double ordinate passing through the focus is called the latus rectum of the conic.
- Centre :** The point which bisects every chord of the conic passing through it is called the centre of the conic.

If the equation of conic is $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, then the coordinates of centre of conic is

$$\left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right) (ab \neq h^2).$$

Focal Distance : The distance from focus to any point on a conic is called focal distance.

- For any conic, if semilatusrectum is ' l ' and the perpendicular from the focus to directrix is ' d ' then

$$\frac{l}{d} = e$$

PARABOLA

Definition : The locus of a point in a plane such that its distance from a fixed point (focus) is always equal to its distance from a fixed straight line (directrix) is called a Parabola.

For the parabola eccentricity (e)=1

Various forms of Parabola. ($a > 0$)

S.No.	Content	I	II	III	IV
	Equation	$y^2 = 4ax$	$y^2 = -4ax$	$x^2 = 4ay$	$x^2 = -4ay$
	Figure				
1	Vertex(A)	(0, 0)	(0, 0)	(0, 0)	(0, 0)
2	Focus (S)	(a, 0)	(-a, 0)	(0, a)	(0, -a)
3	Point of intersection of Axis and directrix (Z)	(-a, 0)	(a, 0)	(0, -a)	(0, a)
4	End points of latus rectum (L,L')	(a, ± 2a)	(-a, ± 2a)	(± 2a, a)	(± 2a, -a)
5	Eqn. of axis	$y = 0$	$y = 0$	$x = 0$	$x = 0$
6	Eqn. of Directrix	$x = -a$	$x = a$	$y = -a$	$y = a$
7	Eqn. of tangent at Vertex	$x = 0$	$x = 0$	$y = 0$	$y = 0$
8	Eqn. of latus recta	$x = a$	$x = -a$	$y = a$	$y = -a$
9	Length of latus rectum (LL')	4a	4a	4a	4a
10	Distance from Focus to Directrix (SZ)	2a	2a	2a	2a
11	SA=AZ	a	a	a	a

Note :

- Length of latusrectum of $ax^2 + bx + cy + n = 0$ is

$$\left| \frac{c}{a} \right| = \left| \frac{\text{Coefficient of } y}{\text{Coefficient of } x^2} \right|$$

- Length of latusrectum of $ay^2 + by + dx + n = 0$ is

$$\left| \frac{d}{a} \right| = \left| \frac{\text{Coefficient of } x}{\text{Coefficient of } y^2} \right|$$

- Let P be a point on the parabola and M, N be its projections on the principal axis and tangent at the vertex. The equation of parabola with latusrectum 4a is $PM^2 = (4a)PN$.

- Let $lx + my + n = 0$ and $mx - ly + k = 0$ be the tangent at vertex and axis of the parabola with latusrectum 4a. The equation of parabola

$$\text{is } (mx - ly + k)^2 = (4a)\sqrt{l^2 + m^2} |lx + my + n|$$

- The equation of the parabola whose axis is parallel to

i) x - axis is of the form $x = ay^2 + by + c$

ii) y - axis is of the form $y = ax^2 + bx + c$

- The parabola $y = ax^2 + bx + c$ is symmetrical

about the line $2ax + b = 0 \left(\frac{dy}{dx} = 0 \right)$.

- The parabola $x = ay^2 + by + c$ is symmetrical

about the line $2ay + b = 0 \left(\frac{dx}{dy} = 0 \right)$

- The focal distance of $P(x_1, y_1)$ on the parabola

i) $y^2 = 4ax$ is $|x_1 + a|$.

ii) $y^2 = -4ax$ is $|x_1 - a|$

iii) $x^2 = 4ay$ is $|y_1 + a|$

iv) $x^2 = -4ay$ is $|y_1 - a|$

Various forms of the Parabola with vertex (h,k):

Parabola $(y - k)^2 = 4a(x - h)$

- Vertex $A(h, k)$
➤ Focus $S(h + a, k)$

- Equation of the directrix is $x = h - a$
➤ Equation of the axis is $y = k$
➤ Equation of the tangent at the vertex is $x = h$
➤ Length of the latusrectum is $LL' = 4a$
➤ Extremities of latusrectum are $L(a + h, k + 2a)$ and $L'(a + h, k - 2a)$.

- Equation of latusrectum is $x = h + a$

Parabola $(x - h)^2 = 4a(y - k)$

- Vertex $A(h, k)$
➤ Focus $S(h, a + k)$
➤ Equation of the directrix is $y = k - a$
➤ Equation of the axis is $x = h$
➤ Equation of the tangent at the vertex is $y = k$
➤ Length of the latusrectum is $LL' = 4a$
➤ Extremities of latusrectum are

$L(h + 2a, k + a)$ and $L'(h - 2a, k + a)$

- Equation of latusrectum is $y = k + a$

Some useful results on double ordinate

- If ' θ ' is angle subtended by double ordinate at

vertex of $y^2 = 4ax$ then its length is $8a \cot \frac{\theta}{2}$.

- The length of double ordinate of the parabola which is at a distance of 'p' units from the vertex is

$4\sqrt{ap}$ where $a = \frac{LLR}{4}$

W.E-1: The length of the double ordinate of the parabola $(y - 3)^2 = 16(x - 2)$ which is at a distance of 9 units from (2,3) is

Sol: Clearly (2,3) is vertex of a parabola where $p=9$, $a=4$.

Length of double ordinate $= 4\sqrt{ap} = 4\sqrt{4 \times 9} = 24$

- The double ordinate of the parabola is perpendicular to the axis of the parabola.

W.E-2: Ends of the double ordinate of $y^2 = 16x$ which is at a distance of 8 units from the vertex is

Sol: Vertex (0,0) axis of the parabola is x-axis

$\therefore$ let one end of double ordinate $= (8, y_1)$

since the point lies on the parabola

$y_1^2 = 16 \times 8 \Rightarrow y_1 = \pm 8\sqrt{2}$

$\therefore$ ends of double ordinate $= (8, 8\sqrt{2}), (8, -8\sqrt{2})$

Parametric Equation of Parabola**Parabola Parametric Equations**

$$y^2 = 4ax \quad (at^2, 2at)$$

$$y^2 = -4ax \quad (-at^2, 2at)$$

$$x^2 = 4ay \quad (2at, at^2)$$

$$x^2 = -4ay \quad (2at, -at^2)$$

➤ **Notation :** $S \equiv y^2 - 4ax$ $S_1 \equiv yy_1 - 2a(x + x_1)$

$$S_{12} \equiv y_1y_2 - 2a(x_1 + x_2) \quad S_{11} \equiv y_1^2 - 4ax_1$$

Position of a point with respect to Parabola

A parabola divides the plane into three regions.

- The region containing the focus, called interior of the parabola.
- The region consisting of the curve itself.
- The region containing the remaining portion of the plane, called the exterior of the parabola.
- The point (x_1, y_1) lies inside or on or outside the

parabola $S=0$ according as $S_{11} \begin{matrix} < \\ = \\ > \end{matrix} 0$.

Equation of Chord

Let $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$ are two points on the parabola $y^2 = 4ax$. Then, the equation of chord is $y(t_1 + t_2) = 2x + 2at_1t_2$

If the chord PQ is a focal chord of the parabola, then $(a, 0)$ must satisfy this equation.

$$0 = 2a + 2at_1t_2 \Rightarrow t_1t_2 = -1.$$

Thus, if t is the parameter for one end of a focal chord, then parameter for other end is $-\frac{1}{t}$ and the coordinates of the end points of a focal chord PQ of the parabola $y^2 = 4ax$ can be taken as

$$P(at^2, 2at) \text{ and } Q\left(\frac{a}{t^2}, \frac{-2a}{t}\right).$$

- Equation of the chord joining the two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ on the parabola $S = y^2 - 4ax = 0$ is $S_1 + S_2 = S_{12}$.

Length of focal chord : Let $P(at^2, 2at)$ be the one end of a focal chord PQ of the parabola $y^2 = 4ax$. Then the length of focal chord is $a\left(t + \frac{1}{t}\right)^2$, where t is the parameter for one end of the chord.

- If l_1 and l_2 are two lengths of focal segments, then,

$$4a = \frac{l_1l_2}{l_1 + l_2} \quad \text{i.e., Length of latusrectum} = 2$$

(Harmonic mean of the focal segments).

Note : The length of focal chord having parameters $P(t_1)$ and $Q(t_2)$ for its end point is $a(t_2 - t_1)^2$.

W.E-3 : The length of focal chord of parabola

$y^2 = 4ax$ makes an angle α with the x-axis, is

Sol : Let $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$ be the end points of a focal chord PQ which makes an angle α with the axis of the parabola. Then,

$$PQ = a(t_2 - t_1)^2 = a[(t_2 + t_1)^2 - 4t_1t_2] \\ = a[(t_2 + t_1)^2 + 4] \quad (\because t_1t_2 = -1) \dots (i)$$

$$\therefore \tan \alpha = \text{slope of } PQ = \frac{2at_2 - 2at_1}{at_2^2 - at_1^2}$$

$$\Rightarrow \tan \alpha = \frac{2}{t_2 + t_1} \Rightarrow t_2 + t_1 = 2 \cot \alpha \dots (ii)$$

On substituting the value of $t_2 + t_1$ from Eq.(ii) in

Eq.(i) we get $PQ = a(4 \cot^2 \alpha + 4) = 4a \operatorname{cosec}^2 \alpha$

Some useful results on focal chord

- Least length of focal chord is $4a$.
- If the tangents and normals at the extremities of a focal chord of a parabola intersect at (x_1, y_1) and (x_2, y_2) respectively then $y_1 = y_2$.
- If the tangents at the ends of the two focal chords of a parabola intersect at (x_1, y_1) and (x_2, y_2) respectively then $x_1 = x_2$.
- If $(at^2, 2at)$ is one end of the focal chord of the parabola $y^2 = 4ax$. Then the other end is $\left(\frac{a}{t^2}, \frac{-2a}{t}\right)$.

- The product of abscissae of the extremities of the focal chord of the parabola $y^2 = 4ax$ is a^2 .
- The product of ordinates of the extremities of the focal chord of the parabola $y^2 = 4ax$ is $-4a^2$.
- If $(x_1, y_1), (x_2, y_2)$ are extremities of a focal chord of the parabola. Then
 - (i) $x_1 \cdot x_2 = a^2$ (ii) $y_1 \cdot y_2 = -4a^2$
- (a) If PQ is focal chord of the parabola $y^2 = 4ax$ and S is the focus then $\frac{1}{SP} + \frac{1}{SQ} = \frac{2}{l}$. Where l is the length of semi latusrectum.
 - (b) $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$ (c) SP, l , SQ are in HP.
 - (d) Semi latusrectum = $\frac{2 \cdot SQ \cdot SP}{SP + SQ}$
- The circle drawn on focal chord of a parabola as diameter touches the directrix.
- The tangent at one end of a focal chord of a parabola is parallel to the normal at the other end.

W.E-4: If $P(2, 1)$ is one end of focal chord PQ of the parabola $x^2 + 2y - 2x - 2 = 0$ then the slope of the normal at Q is :

Sol: Given parabola is $x^2 + 2y - 2x - 2 = 0$

If differentiate w.r.t.x

$$2x + 2 \frac{dy}{dx} - 2 = 0 \Rightarrow \frac{dy}{dx} = 1 - x$$

Slope of the tangent at $P = 1 - 2 = -1$

$\therefore$ Slope of normal at Q = -1.

Intersection of Line and Parabola

Let the parabola be $y^2 = 4ax$ (i)

and the given line be $y = mx + c$ (ii)

On eliminating x from Eqs.(i) and (ii), then

$$y^2 = 4a \left(\frac{y-c}{m} \right) \Rightarrow my^2 - 4ay + 4ac = 0 \text{ ... (iii)}$$

It is a quadratic equation in y. Discriminant,

$$D = (-4a)^2 - 4(4ac)m = 16a(a - cm)$$

- If $D > 0$ i.e., $a > cm$, then line intersect the parabola at two distinct points.
- If $D = 0$ i.e., $a = cm$ then line intersect the parabola at two coincident points i.e., at one point.
- If $D < 0$ i.e., $a < cm$, then line neither touch nor intersect the parabola.

Equation of Tangent & Normal

- Equation of the tangent to $y^2 = 4ax$ at (x_1, y_1) is $S_1 = 0$ i.e., $yy_1 = 2a(x + x_1)$.
- Equation of the tangent to $y^2 = 4ax$ at the point 't' is $yt = x + at^2$.
- Equation of the tangent to $y^2 = 4ax$ having slope m is $y = mx + \frac{a}{m}$.
- Equation of the normal to $y^2 = 4ax$ at (x_1, y_1) is $y - y_1 = \frac{-y_1}{2a}(x - x_1)$
- Equation of the normal to $y^2 = 4ax$ at the point 't' is $y + xt = 2at + at^3$.
- Equation of the normal to $y^2 = 4ax$ having slope m is $y = mx - 2am - am^3$.

Some useful results on Tangents

- The condition for the line $y = mx + c$ to touch $y^2 = 4ax$ is $c = \frac{a}{m}$ and the point of contact is $\left(\frac{a}{m^2}, \frac{2a}{m} \right)$. And equation of tangent in slope form is $y = mx + \frac{a}{m}$.
- The condition for $y = mx + c$ to touch $x^2 = 4ay$ is $c = -am^2$. and point of contact is $(2am, am^2)$.
- The condition for $lx + my + n = 0$ to touch $y^2 = 4ax$ is $am^2 = ln$ and the point of contact is $\left(\frac{n}{l}, \frac{-2am}{l} \right)$.
- The condition for $lx + my + n = 0$ to touch $x^2 = 4ay$ is $al^2 = mn$ and the point of contact is $\left(\frac{-2al}{m}, \frac{n}{m} \right)$.
- The condition that the line $y = mx + c$ may be tangent to the parabola $y^2 = 4a(x + a)$ is $c = am + \frac{a}{m}$.

- Equation of the tangent in slope form to the parabola

$$y^2 = 4a(x+a) \text{ is } y = m(x+a) + \frac{a}{m}.$$

- Equation of the tangent to the parabola

$$(y-k)^2 = 4a(x-h) \text{ is } (y-k) = m(x-h) + \frac{a}{m}$$

and Point of contact is $\left(h + \frac{a}{m^2}, k + \frac{2a}{m}\right)$

- The condition that the line $lx + my + n = 0$ is a tangent to the parabola $y^2 = 4a(x-a)$ is $a(m^2 - l^2) = ln$.

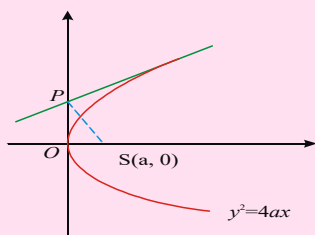
- The length of the perpendicular from focus S of the parabola $y^2 = 4ax$ on the tangent at P is $\sqrt{OS \cdot SP}$.

- If the line $lx + my + n = 0$ intersects the parabola $y^2 = 4ax$ at A and B. then the point of intersection of tangents at A and B is $\left(\frac{n}{l}, \frac{-2am}{l}\right)$.

- If two tangents at the points t_1 and t_2 on the parabola are perpendicular then $t_1 t_2 = -1$.

W.E-5: Show that Locus of foot of the perpendicular drawn from the focus on any tangent is the tangent at the vertex of the parabola.

Sol: for $y^2 = 4ax$, $x = 0$ is the tangent at the vertex



Given parabola $y^2 = 4ax$

Equation of any tangent to the parabola is

$$y = mx + \frac{a}{m} \Rightarrow m^2 x - my + a = 0 \rightarrow (1)$$

Equation of any line perpendicular to (1) and passing through $S(a, 0)$ is $m(x-a) + m^2 y = 0$

$$\Rightarrow x + my = a \text{ Solving (1) \& (2) we get } P\left(0, \frac{a}{m}\right)$$

Which is the foot of the perpendicular from $S(a, 0)$ on the tangent.

Hence the locus of P is, $x = 0$ which is the tangent at the vertex.

- If SP and SQ are the distances of two points P and Q on the parabola $y^2 = 4ax$ from focus S. T is the point of intersection of tangent at P and Q then $ST^2 = SP \cdot SQ$ and TP, TQ subtend equal angles at S.

- The point of intersection of the tangents at t_1, t_2 on the parabola $y^2 = 4ax$ is $[at_1 t_2, a(t_1 + t_2)]$.

- The ordinate of the point of intersection of the two tangents to a parabola is the arithmetic mean between the ordinates of the points of contact.

- The tangents at the ends of a focal chord of the parabola meet on the directrix at right angle.

- Angle between tangents at two points $P(at_1^2, 2at_1)$ and $Q(at_2^2, 2at_2)$ on the parabola

$$y^2 = 4ax \text{ is } \theta = \tan^{-1} \left| \frac{t_2 - t_1}{1 + t_1 t_2} \right|$$

- The angle between the tangents to the parabola drawn at the ends of the latusrectum is $\pi/2$.

- The perpendicular tangents to the parabola $y^2 = 4ax$ intersect on the directrix of the parabola.

Common tangents

- Equation of common tangent to the parabolas $y^2 = 4ax$ and $x^2 = 4by$ is

$$xa^{1/3} + yb^{1/3} + a^{2/3}b^{2/3} = 0$$

- The maximum number of common normals to $y^2 = 4ax$ and $x^2 = 4by$ may be 5 only.

- The equation of the common tangent to the circle $x^2 + y^2 = 2a^2$ and a parabola $y^2 = 8ax$ is $y = \pm(x + 2a)$.

Results on normals to Parabola

- Equation of the normal to $y^2 = 4ax$ at the point 't' is $tx + y = 2at + at^3$, $\sum t_i = 0$,

$$\sum t_1 t_2 = \frac{2a - x_1}{a}, \quad t_1 t_2 t_3 = \frac{y_1}{a}$$

- The equation of the normal to the parabola $y^2 = 4ax$ at $(at^2, 2at)$ is $y + xt = 2at + at^3$. If this line passes through (x_1, y_1) then

$$y_1 + x_1 t = 2at + at^3 \text{ becomes}$$

$$at^3 + t(2a - x_1)t - y_1 = 0.$$

This is a cubic equation in 't' and has atmost three roots. Hence the number of normals through a given point (x_1, y_1) to the parabola $y^2 = 4ax$ is either 1 or 2 or 3 accordingly as the number of distinct real roots of the cubic equation

$$at^3 + (2a - x_1)t - y_1 = 0 \text{ is 1 or 2 or 3.}$$

➤ Write $H = \frac{2a - x_1}{3a}$, $G = \frac{-y_1}{a}$ and $\Delta = G^2 + 4H^3$

If $x_1 = 2a$ and $y_1 = 0$, then the number of normals = 1

➤ Assume either $x_1 \neq 2a$ or $y_1 \neq 0$.

➤ If $\Delta > 0$, then the number of normal is 1

If $\Delta = 0$, then the number of normals is 2

If $\Delta < 0$, then the number of normals is 3

➤ The condition for $lx + my + n = 0$ to be a normal to $y^2 = 4ax$ is $al^3 + 2alm^2 + m^2n = 0$

➤ i) Equation of the normal to $y^2 = 4ax$ having slope m is $y = mx - 2am - am^3$ and the foot of the normal is $(am^2, -2am)$.

ii) $y = mx + c$ is a normal to the parabola $y^2 = 4ax$ then $c = -2am - am^3$.

➤ Slope of the tangent and normal at the point 't' to $y^2 = 4ax$ are $\frac{1}{t}, -t$.

➤ The point of intersection of normals drawn at t_1 and t_2 to $y^2 = 4ax$ is

$$\left[2a + a\{(t_1 + t_2)^2 - t_1t_2\}, at_1t_2(t_1 + t_2) \right].$$

➤ If the normal at $(at_1^2, 2at_1)$ cuts the parabola again

at $(at_2^2, 2at_2)$ then $t_2 = -t_1 - \frac{2}{t_1}$.

➤ If the normals at the points t_1 and t_2 on the parabola intersect on the parabola at the point t_3 then

(a) $t_1t_2 = 2$ (b) $t_1 + t_2 = -t_3$.

➤ If the normal at $P(at^2, 2at)$ to the parabola meets the parabola again at Q, then there will be one more point on the parabola at which the normal will pass through Q, the co-ordinate of that point is

$$\left(\frac{4a}{t^2}, \frac{4a}{t} \right).$$

➤ From a point which lies inside the parabola three normals can be drawn to the parabola. The sum of the ordinates of the feet of the three normals is zero.

➤ If the normal at P meets the axis of the parabola $y^2 = 4ax$ at G then $SP = SG$.

➤ The length of the normal chord drawn at the point 't' on the parabola $y^2 = 4ax$ is $\frac{4a(1+t^2)^{3/2}}{t^2}$.

➤ The length of the normal chord at one end of the latusrectum of $y^2 = 4ax$ is $8\sqrt{2a}$.

➤ If the normal at the point 't' on the parabola $y^2 = 4ax$ subtends a right angle.

a) at its focus then $t = \pm 2$

b) at its vertex then $t = \pm\sqrt{2}$.

W.E-6: The normal chord of the parabola $y^2 = 4ax$ subtends a right angle at the focus. Then the length of chord is

Sol: $t = \pm 2$

$$\text{Length of normal chord} = \frac{4a(1+t^2)^{3/2}}{t^2} = 5\sqrt{5}.a.$$

W.E-7: The normal chord of the parabola $y^2 = 4ax$ subtends a right angle at the focus. Then the end point of the chord is

Sol: Let t be the end point of the normal chord. Since normal chord subtends a right angle at focus

$$\Rightarrow t = \pm 2. \text{ end points of the chord} = (at^2, 2at) = (4a, 4a) \text{ or } (4a, -4a)$$

W.E-8: The normal chord of the parabola $y^2 = 4ax$ subtends a right angle at the vertex. Then the length of chord is

Sol: $t = \pm\sqrt{2}$; length of the normal chord

$$= \frac{4a(1+t^2)^{3/2}}{t^2} = 6\sqrt{3}.a$$

W.E-9: A normal chord of a parabola at 't' point subtends a right angle at the vertex. Then the chord makes an angle with axis of the parabola is...

Sol: Take the normal at the point 't'

$$\Rightarrow t = \pm\sqrt{2}$$

Slope of the normal at the point P(t) = -t

$$\Rightarrow \tan \theta = t \Rightarrow \tan \theta = \pm\sqrt{2} \Rightarrow \theta = \tan^{-1}(\pm\sqrt{2})$$

W.E-10: A normal chord of the parabola $y^2 = 4ax$ makes an angle 45° with the axis of the parabola. Then its length is

Sol: Normal $y = mx - 2am - am^3 \Rightarrow y = x - 3a$

$$\therefore \text{Its length} = \frac{4}{m^2} \cdot \sqrt{a(a - mc)(1 + m^2)}$$

Where $m=1, c=-3a$

- If two normals at the points t_1 and t_2 on the parabola are perpendicular then $t_1 t_2 = -1$.
- If a tangent at t_1 is parallel to the normal at t_2 point on the parabola then $t_1 t_2 = -1$.
- The normal and the tangent at a point P on the parabola meets the axis in G and T respectively then
 - a) ST, SG, SP are in G.P
 - b) Triangle SPG is an equilateral.
- An equilateral triangle incised in the parabola $y^2 = 4ax$ with one vertex coincide with the vertex of the parabola then side of the triangle is $8a\sqrt{3}$.
- c) Length of the side of the triangle $= 4a$.

Pair of tangents and their chord of contact

- Equation of the chord of contact of (x_1, y_1) with respect to $y^2 = 4ax$ is $S_1 = 0$ i.e., $yy_1 = 2a(x + x_1)$.
- Area of the triangle formed by the tangents from (x_1, y_1) to the parabola $y^2 = 4ax$ and its chord of contact is $\frac{(y_1^2 - 4ax_1)^{3/2}}{2a} = \frac{S_{11}^{3/2}}{2a}$
- The length of chord of contact of tangents drawn from (x_1, y_1) to the parabola $y^2 = 4ax$ is $\frac{\sqrt{(y_1^2 - 4ax_1)(y_1^2 + 4a^2)}}{a}$.
- From an external point two tangents can be drawn to a parabola. The slopes of the two tangents to $y^2 = 4ax$ passing through (x_1, y_1) are given by the equation $m^2 x_1 - m y_1 + a = 0$.
- If m_1, m_2 be the slopes of the two tangents drawn from (x_1, y_1) to $y^2 = 4ax$ then $m_1 + m_2 = \frac{y_1}{x_1}$,

$$m_1 m_2 = \frac{a}{x_1}.$$

- If θ is the angle between the pair of tangents drawn from (x_1, y_1) to $y^2 = 4ax$ then

$$\tan \theta = \frac{\sqrt{y_1^2 - 4ax_1}}{x_1 + a} \text{ or } \frac{\sqrt{S_{11}}}{x_1 + a}.$$

- The locus of point of intersection of the two tangents to $y^2 = 4ax$ included at a constant angle α is $(x + a)^2 \tan^2 \alpha = y^2 - 4ax$.
- Locus of point of intersection of perpendicular tangents to a parabola is its directrix. i.e., $x + a = 0$.
- Equation of pair of tangents to the parabola $S = 0$ from $P(x_1, y_1)$ is $S_1^2 = S \cdot S_{11}$
- Slopes of tangents from (x_1, y_1) to the parabola

$$y^2 = 4ax \text{ are } \frac{y_1 \pm \sqrt{y_1^2 - 4ax_1}}{2x_1}$$

- If the tangents from P to the parabola $y^2 = 4a(x + a)$ and $y^2 = 4b(x + b)$ are at right angles. Then the locus of P is $x + a + b = 0$.

Condition	No. of tangents can be drawn to the parabola	No. of normals can be drawn to the parabola (If m is real)
$S_{11} = 0$	1	2
$S_{11} < 0$	0	3
$S_{11} > 0$	2	1

- P is a point on the parabola whose focus is S and PN is the perpendicular drawn from P to the directrix, then the tangent at P is the internal bisector of $\angle SPN$ and normal is the external bisector of $\angle SPN$.

Equation of a chord bisect at given point

- Equation of the chord of the parabola $y^2 = 4ax$ with mid point (x_1, y_1) is $S_1 = S_{11}$ i.e., $yy_1 - 2ax = y_1^2 - 2ax_1$.
- If the line $y = mx + c$ intersect the parabola $y^2 = 4ax$ in the points A and B then:
- (1) The length of the chord AB is

$$\frac{4}{m^2} \sqrt{a(a - mc)(1 + m^2)}$$

(2) The mid point of the chord AB is $\left(\frac{2a-mc}{m^2}, \frac{2a}{m}\right)$.

➤ The length of chord joining the points t_1 and t_2 on the parabola $y^2 = 4ax$ is $a|t_1 - t_2| \cdot \sqrt{(t_1 + t_2)^2 + 4}$

➤ The length of the chord of $y^2 = 4ax$ having (x_1, y_1) on its mid point is $\frac{1}{a} \sqrt{-(y_1^2 - 4ax_1)(y_1^2 + 4a^2)}$

➤ The locus of the chord of the parabola $y^2 = 4ax$ passing through vertex and making an angle ' θ ' with the axis is $4a \cos \theta \cdot \operatorname{cosec}^2 \theta$.

W.E-11: The locus of middle points of all chords of the parabola $y^2 = 4ax$ passing through the vertex of the parabola is

Sol: Use $S_1 = S_{11}$ passes through $(0,0)$

W.E-12: The locus of mid points of the chords of the parabola $y^2 = 4(x+1)$ which are parallel to $3x = 4y$ is

Sol: $S_1 = S_{11}$ use slope $m = \frac{3}{4}$

W.E-13: Locus of mid points of the chords of the parabola $y^2 = 4ax$ which touch the circle $x^2 + y^2 = a^2$ is

Sol: $S_1 = S_{11}$ is tangent to $x^2 + y^2 = a^2$

W.E-14: The locus of mid points of the chords of the parabola $y^2 = 4ax$ passing through the foot of the directrix is.

Sol: $S_1 = S_{11}$ passes through $Z(-a, 0)$

W.E-15: The locus of mid points of normal chords of the parabola $y^2 = 4ax$ is

Sol: Let $P(x_1, y_1)$ be a midpoint of the chord then the equation of the chord is $S_1 = S_{11}$ apply normal condition $al^3 + 2alm^2 + m^2n = 0$.

$$a(2a)^3 + 2a(2a)(y_1)^2 + y_1^2(y_1^2 - 2ax_1) = 0$$

Locus of mid-point is

$$8a^4 + 4ay^2 + y^2(y^2 - 2ax) = 0$$

W.E-16: If the locus of mid points of the chords of the parabola $y^2 = 4ax$ which passes through a fixed point (h,k) is also a parabola then its length of latusrectum is

Sol: Use $S_1 = S_{11}$; $yy_1 - 2a(x + x_1) = y_1^2 - 4ax_1$ and it passes through (h,k)

$$ky_1 - 2a(h + x_1) = y_1^2 + 4ax_1$$

Locus of middle point is $y^2 - 4ax = ky - 2a(h + x)$

$$y^2 = 2ax + ky - 2ah$$

$$\therefore L.L.R = \left| \frac{\text{coeff of } x}{\text{coeff of } y^2} \right| = \left| \frac{2a}{1} \right| = 2a$$

Miscellaneous :

➤ The orthocentre of the triangle formed by the tangents at t_1, t_2, t_3 to the parabola $y^2 = 4ax$ is $[-a, a(t_1 + t_2 + t_3 + t_1t_2t_3)]$.

➤ Orthocentre of the triangle formed by any three tangents to the parabola lies on the directrix of the parabola.

➤ The centroid of the triangle formed by the feet of the three normals of the parabola lies on the axis of the parabola.

➤ The length of sub tangent at any point $P(x_1, y_1)$ on $y^2 = 4ax$ is $|2x_1|$.

➤ P is a point on the parabola $y^2 = 4ax$ with focus S. If the tangent and normal at P meets the axis at T and N then $ST = SP = SN = |x_1 + a|$.

➤ Area of the triangle inscribed in parabola $y^2 = 4ax$ is $\frac{1}{8a} |(y_1 - y_2)(y_2 - y_3)(y_3 - y_1)|$ sq. units where y_1, y_2 & y_3 are ordinates of angular points.

➤ Area of the triangle formed by the tangents at three points whose the ordinates y_1, y_2, y_3 on $y^2 = 4ax$

$$\text{is } \frac{1}{16a} |(y_1 - y_2)(y_2 - y_3)(y_3 - y_1)| \text{ sq. units.}$$

➤ The point where a diameter of a parabola meets the parabola is called the vertex of the diameter.

➤ The tangent to a parabola at the vertex of a diameter is parallel to the chords bisected by that diameter.

➤ A circle meets a parabola in general in four points, the algebraic sum of whose ordinates is zero.

➤ The angle between the two parabolas $y^2 = 4ax$,

$$x^2 = 4by \text{ is } \tan^{-1} \left[\frac{3a^{1/3} \cdot b^{1/3}}{2(a^{2/3} + b^{2/3})} \right].$$

- The ordinate of any point on the parabola is a mean proportional between the latusrectum and the abscissa.
- The circumcircle of the triangle formed by any three tangents to parabola passes through its focus.
- The point on the parabola which is nearest to the directrix is its vertex.
- The point on the parabola which is nearest to the focus is vertex.
- The tangents and normals at the ends of latusrectum of parabola $y^2 = 4ax$ forms a square whose side is $2\sqrt{2}a$ and area is $8a^2$.

**C.U.Q**

1. Number of focal chords of the parabola $y^2 = 9x$ whose length is less than 9 is
1) 2 2) 5 3) 1 4) 0
2. The foci of the parabolas $y^2 = 4x$, $x^2 = 4y$, $y^2 + 4x = 0$, $x^2 + 4y = 0$ taken in order are the vertices of a
1) rectangle 2) parallelogram
3) rhombus 4) square
3. In which quadrants the parabola $x^2 = -4y$ does not exist
1) I, II 2) II, III
3) III, IV 4) I, IV
4. Locus of the point equidistant from (0, -1) and the line $y=1$ is
1) a parabola with vertex (0, 0)
2) a parabola with focus (0, 1)
3) a parabola with directrix $y=-1$
4) a parabola with axis $y=0$
5. The point on the parabola which is nearest to directrix is
1) End of latusrectum 2) Focus
3) Vertex 4) Centre.
6. The locus of the centre of a circle passing through a point and touching a line is
1) a straight line 2) an ellipse
3) a parabola 4) a hyperbola
7. When a circle with centre lying on the focus of a parabola touches its directrix then the radius of the circle is
1) $1/4$ (length of latusrectum of the parabola)
2) $1/2$ (length of latusrectum of the parabola)
3) $3/4$ (length of latusrectum of the parabola)
4) $2/3$ (length of latusrectum of the parabola)
8. The number of tangents $y^2 = 4ax$ which are parallel to Y-axis
1) 1 2) 3 3) 0 4) Infinite
9. The two parabolas $y^2 = 4x$ and $x^2 = 4y$ intersect at a point P, whose abscissa is not zero, such that
1) they both touch each other at P
2) they cut at right angles at P
3) the tangents to each curve at P make complementary angles with x-axis
4) None
10. We are given with two ends of the latus-rectum of a parabola, then the maximum number of parabolas, which can be drawn is
1) 0 2) 1 3) 2 4) infinite
11. $y = \sqrt{x}$ represents
1) part of ellipse 2) semi parabola
3) parabola 4) circle
12. The point on $y^2 = 4ax$ nearest to the focus has its abscissa equal to
1) $-a$ 2) a 3) $a/2$ 4) 0
13. The abscissa of the orthocentre of the triangle formed by the lines $y = m_1x + \frac{b}{m_1}$, $y = m_2x + \frac{b}{m_2}$, $y = m_3x + \frac{b}{m_3}$ is
1) b 2) -b 3) 2b 4) -2b
14. The circumcircle of the triangle formed by any three tangents to a parabola passes through
1) vertex 2) the ends of latusrectum
3) the focus 4) the mid point of focus and vertex
15. A ray of light moving parallel to the x-axis gets reflected from a parabolic mirror whose equation is $(y-2)^2 = 4(x+1)$. The point on the axis of parabola through which the ray after reflection passes is
1) (0,2) 2) (0,3) 3) (0,4) 4) (0,1)
16. The tangents and normals at the ends of the latus rectum of a parabola form
1) cyclic quadrilateral 2) rectangle
3) a square 4) circle

C.U.Q - KEY

- 1) 4 2) 4 3) 1 4) 1 5) 3 6) 3
7) 2 8) 1 9) 3 10) 3 11) 2 12) 4
13) 2 14) 3 15) 1 16) 3

C.U.Q - HINTS

- Least length of the focal chord is $4a$
- The tangents at the ends of the latus rectum to $y^2 = 4ax$ i.e., at $L(a, 2a)$ and $L'(a, -2a)$ are $x - y + a = 0$ and $x + y + a = 0$ respectively since, they are $\perp$ $\therefore$ their normals are also $\perp$. Also, these tangents intersect at $P(-a, 0)$ $\therefore PL = PL'$ Hence, they form a square.
- $y < 0$ graph lies in III and IV
- Equation of the directrix is $x + a = 0$
Let $p(at^2, 2at)$ the any point on the parabola,
distance $= at^2 + a \Rightarrow f'(t) = 0 \Rightarrow t = 0$
- The locus which is equidistance from a fixed point and line is a parabola
- $s(a, 0)$ the radius of the circle is $SZ = 2a$
- There exist one tangent at the vertex
- The point of intersection is $(4, 4)$. Let slopes of tangents at that point be m_1, m_2 . Then $m_1 m_2 = 1$
- L_1 and L_2 are given ends of latusrectum S bisects line $L_2 L_1$. $ASA^\perp$ is the perpendicular bisector of line $L_2 L_1$. where $AS = \frac{1}{4} L_1 L_2 = A^\perp S$
Hence two parabolas are possible.
- Semi parabola
- The point on $y^2 = 4ax$, which is nearest to focus is vertex.
- $x = -b$
- The circumcircle of the triangle formed by any three tangents to a parabola passes through the focus of the parabola.
- Any ray parallel to the axis of parabola passes through the focus after reflection.
- tangent at vertex is perpendicular to the axis $\overline{AS}$



LEVEL-I (CW)

FOCUS, VERTEX, DIRECTRIX, LATUSRECTUM, AXIS, FOCAL CHORD, FOCAL DISTANCE, DOUBLE ORDINATE

- If a parabolic reflector is 20 cm in diameter and 5 cm deep, then the distance from vertex to focus is
1) 5cm 2) 10cm 3) 15cm 4) 20cm
- The focus and vertex of a parabola are $(4, 5)$ and $(3, 6)$. The equation of axis is
1) $2x - y + 3 = 0$ 2) $2x - y = 0$
3) $2x + y - 13 = 0$ 4) $x + y - 9 = 0$
- If the focus and directrix of a parabola are $(3, -4)$ and $x + y + 7 = 0$ then its length of latusrectum is
1) $3\sqrt{2}$ 2) $8\sqrt{2}$ 3) $10\sqrt{2}$ 4) $6\sqrt{2}$
- The focus and directrix of a parabola are $(1, 2)$ and $2x - 3y + 1 = 0$. Then the equation of the tangent at the vertex is
1) $4x - 6y + 5 = 0$ 2) $4x - 6y + 9 = 0$
3) $4x - 6y + 11 = 0$ 4) $4x - 6y + 7 = 0$
- The focus and directrix of a parabola are $(1, -1)$ and $x + y + 3 = 0$. Its vertex is
1) $(\frac{7}{4}, \frac{1}{4})$ 2) $(\frac{1}{2}, \frac{-7}{4})$ 3) $(\frac{1}{4}, \frac{-7}{4})$ 4) $(\frac{1}{2}, \frac{5}{2})$
- The focal distance of a point on the parabola $y^2 = 8x$ is 10. Its coordinates are
1) $(2, \pm 2)$ 2) $(3, \pm 3)$ 3) $(5, \pm 5)$ 4) $(8, \pm 8)$
- The angle subtended by double ordinate of length $8a$ of the parabola $y^2 = 4ax$ at its vertex is
1. $\frac{\pi}{3}$ 2. $\frac{\pi}{4}$ 3. $\frac{\pi}{2}$ 4. $\frac{\pi}{6}$
- The length of the latusrectum of the parabola $2y^2 + 3y + 4x - 2 = 0$ is
1) $\frac{3}{2}$ 2) $\frac{1}{3}$ 3) 2 4) 6
- Equation of the axis of the parabola $y^2 + 6y - 2x + 5 = 0$ is
1) $y + 3 = 0$ 2) $2y + 3 = 0$ 3) $y + 2 = 0$ 4) $y - 2 = 0$
- Vertex of the parabola $2y^2 + 3y + 4x - 2 = 0$ is
1) $(\frac{25}{32}, \frac{-7}{4})$ 2) $(\frac{25}{32}, \frac{-3}{4})$ 3) $(\frac{15}{32}, \frac{7}{4})$ 4) $(\frac{17}{32}, \frac{-3}{4})$

11. Focus of the parabola $4x^2 - 12x + 8y + 13 = 0$ is

- 1) $\left(\frac{3}{2}, -2\right)$ 2) $\left(\frac{3}{2}, -5\right)$ 3) $\left(\frac{3}{2}, -3\right)$ 4) $\left(\frac{3}{2}, -1\right)$

12. Equation of the directrix of the parabola

$$y^2 = 5x - 4y - 9 \text{ is}$$

- 1) $2x + 1 = 0$ 2) $4x + 1 = 0$
3) $6x + 1 = 0$ 4) $4x - 1 = 0$

13. The ends of latusrectum of a parabola are

$(6, 7), (6, -1)$. Its vertex is

- 1) $(8, 3)$ 2) $(-4, 3)$ 3) $(10, 3)$ 4) $(2, 3)$

14. AB is a focal chord of the parabola $y^2 = 4ax$. If $A = (4a, 4a)$ then B =

- 1) $\left(\frac{a}{2}, \frac{-a}{4}\right)$ 2) $\left(\frac{a}{4}, -a\right)$ 3) $\left(\frac{a}{2}, \frac{-a}{2}\right)$ 4) $\left(\frac{a}{4}, -4a\right)$

15. One extremity of a focal chord of $y^2 = 16x$ is $A(1, 4)$. Then the length of the focal chord at A is

- 1) $\frac{25}{4}$ 2) $\frac{25}{2}$ 3) $\frac{15}{2}$ 4) 25

16. PSQ is a focal chord of the parabola $y^2 = 16x$.

If $P = (1, 4)$ then $\frac{SP}{SQ} =$

- 1) $\frac{3}{4}$ 2) $\frac{2}{3}$ 3) $\frac{1}{9}$ 4) $\frac{1}{4}$

17. The axis of symmetry of the conic

$$y = ax^2 + bx + c \text{ is}$$

- 1) $y = 0$ 2) $x = 0$ 3) $b + 2ax = 0$ 4) $y = ax$

18. Equation of the focal chord of the parabola

$$y^2 = 4x \text{ inclined an angle } \frac{\pi}{4} \text{ with the x axis is}$$

- 1) $x + y - 5 = 0$ 2) $x - y + 2 = 0$
3) $x - y + 4 = 0$ 4) $x - y - 1 = 0$

19. Circles are drawn with diameter being any focal chord of the parabola $y^2 - 4x - y - 4 = 0$ will always touch the fixed line then its equation is

- 1) $16x - 33 = 0$ 2) $16x + 33 = 0$
3) $8x - 33 = 0$ 4) $8x + 33 = 0$

EQUATION OF PARABOLA/CONIC

20. The equation of the parabola whose vertex and focus lie on the axis of x at a distance 'a' and 'a₁' from origin respectively is

1) $y^2 = 4(a_1 - a)x$

2) $y^2 = 4(a_1 - a)(x - a)$

3) $y^2 = 4(a_1 - a)(x - a_1)$

4) $y^2 = 4(a_1 + a)x$

21. The length of the latusrectum of a conic is 5. Its focus is $(-1, 1)$ and its directrix is $3x - 4y + 2 = 0$ then the conic is

- 1) a parabola 2) an ellipse
3) a hyperbola 4) a R.H.

22. Which of the following equations represents a parabola

1) $(x - y)^3 = 3$ 2) $\frac{x}{y} - \frac{y}{x} = 0$

3) $\frac{x}{y} + \frac{4}{x} = 0$ 4) $(x + y)^2 + 3 = 0$

23. Equation of the parabola whose vertex is the origin, axis along the x axis and which passes through the point $(-2, 4)$ is

1) $y^2 = -8x$ 2) $y^2 = -12x$

3) $y^2 = 8x$ 4) $x^2 = y$

24. Equation of the parabola with focus $(3, -4)$ and directrix $x + y + 7 = 0$ is

1) $x^2 + y^2 - 2xy - 26x + 2y + 1 = 0$

2) $x^2 + y^2 - 2xy - 26x + 14y - 3 = 0$

3) $x^2 + y^2 - 2xy - 26x - 14y + 3 = 0$

4) $x^2 + y^2 - 2xy - 26x - 2y + 5 = 0$

25. Equation of the parabola having focus $(3, 2)$ and vertex $(-1, 2)$ is

1) $(x + 1)^2 = 16(y - 2)$ 2) $(x - 1)^2 = 16(y + 2)$

3) $(y - 2)^2 = 16(x + 1)$ 4) $(y + 2)^2 = 16(x - 1)$

26. Equation of the parabola whose axis is horizontal and passing through the points $(-2, 1), (1, 2), (-1, 3)$ is

1) $5y^2 - 2x + 10y + 20 = 0$ 2) $5y^2 + 2x - 21y + 20 = 0$

3) $5y^2 + 2x - 21y + 40 = 0$ 4) $3y^2 - 9x + 10y - 15 = 0$

27. Equation of a parabola whose vertex is $(-3, 4)$, axis is parallel to the y axis and the latusrectum 12 is

1) $(y - 4)^2 = 12(x + 3)$ 2) $(y + 4)^2 = 12(x - 3)$

3) $(x + 3)^2 = 12(y - 4)$ 4) $(x - 3)^2 = 12(y + 4)$

28. Equation of a parabola whose vertex is (2,-3), axis is parallel to the x-axis and latusrectum 8 is

- 1) $(y+3)^2 = 8(x+2)$ 2) $(y+3)^2 = 16(x-2)$
3) $(y+3)^2 = 8(x-2)$ 4) $(y+3)^2 = 32(x-2)$

29. A variable circle passes through the fixed point (2,0) and touches the y-axis. Then the locus of its centre is

- 1) a parabola 2) a circle
3) an ellipse 4) a hyperbola

TANGENT AND ITS APPLICATIONS

30. Focal chord of $y^2 = 16x$ is a tangent to $(x-6)^2 + y^2 = 2$ then possible values of the slopes are

- 1) 1, -1 2) -2, 2 3) -2, 1/2 4) 2, -(1/2)

31. The tangent to the curve $y = x^2 + 6$ at a point (1,7) touches the circle

$x^2 + y^2 + 16x + 12y + c = 0$ at a point Q then the coordinates of Q are

- 1) (-6, -11) 2) (-9, -13) 3) (-10, -15) 4) (-6, -7)

32. Equation of the tangent to $y^2 = 6x$ at the positive end of the latusrectum is

- 1) $2x - 2y + 3 = 0$ 2) $2x + 2y + 3 = 0$
3) $y = 4$ 4) $y + 4 = 0$

33. Equation of the tangent to $y^2 = 8x$ inclined at an angle 30° to the axis is

- 1) $x + \sqrt{3}y + 6 = 0$ 2) $x - \sqrt{3}y + 6 = 0$
3) $\sqrt{3}x + y + 6 = 0$ 4) $\sqrt{3}x - y + 6 = 0$

34. A tangent to $y^2 = 7x$ is equally inclined with the coordinate axes. Then the area of the triangle formed by the tangent with the coordinate axes is

- 1) 25/16 2) 49/32 3) 36/25 4) 49/36

35. If the line $2x + 3y + k = 0$ touches the parabola $x^2 = 108y$ then k =

- 1) 27 2) 18 3) 24 4) 36

36. Equation of the tangent to $y^2 = 4a(x+a)$ having slope 1 is

- 1) $x - y + 2a = 0$ 2) $x + y + a = 0$
3) $x + y - 4a = 0$ 4) $x - y - 2a = 0$

37. Equation of the tangent at the point $t = -3$ to the parabola $y^2 = 3x$ is

- 1) $12x + 4y + 27 = 0$ 2) $4x + 12y - 27 = 0$
3) $4x + 12y + 27 = 0$ 4) $12x + 4y - 27 = 0$

38. Number of tangents drawn from (-2, -3) to the parabola $2y^2 = 9x$ is

- 1) 3 2) 0 3) 1 4) 2

39. The slopes of the two tangents drawn from $(\frac{3}{2}, 5)$ to $y^2 = 6x$ are

- 1) $3, \frac{1}{3}$ 2) $5, \frac{1}{5}$ 3) 7, 3 4) 5, 2

40. If two tangents are drawn from the point (-2, -1) to the parabola $y^2 = 4x$. If α is the angle between these tangents then $\tan \alpha$

- 1) 2 2) 3 3) 4 4) 5

41. A tangent to $y^2 = 5x$ is parallel to the line $y = 4x + 1$. Then the point of contact is

- 1) (5, 5) 2) $(\frac{5}{64}, \frac{5}{8})$ 3) $(\frac{5}{8}, \frac{5}{64})$ 4) (2, 7)

42. For all values of m the line $y = m(x-a) - \frac{a}{m}$ touches the parabola

- 1) $y^2 = 4a(x+a)$ 2) $y^2 = 4a(x-a)$
3) $y^2 = -4a(x+a)$ 4) $y^2 = -4a(x-a)$

43. If the tangents to the parabola $y^2 = 4ax$ at (x_1, y_1) and (x_2, y_2) meet on the axis then

- 1) $x_1 = -x_2$ 2) $x_1 = x_2$ 3) $y_1 = y_2$ 4) $2y_1 = -y_2$

44. If the tangents at t_1, t_2, t_3 on $y^2 = 4ax$ make angles $30^\circ, 45^\circ, 60^\circ$ with the axis then t_1, t_2, t_3 are in

- 1) A.P. 2) G.P. 3) H.P. 4) A.G.P.

45. Two tangents to the parabola $y^2 = 4ax$ make angles θ_1, θ_2 with the x-axis. Then the locus of their point of intersection if $\cot \theta_1 + \cot \theta_2 = c$ is

- 1) $x = ac$ 2) $y^2 - 2ax = cx^2$ 3) $y = ca$ 4) $y = cx$

46. If y_1 and y_2 are the ordinates of two points P and Q on a parabola and y_3 is the ordinate of the point of intersection of the tangents at P and Q then

- 1) y_1, y_2, y_3 are in A.P. 2) y_1, y_3, y_2 are in A.P.
3) y_1, y_2, y_3 are in G.P. 4) y_1, y_3, y_2 are in G.P.

47. The point of intersection of the two tangents at the ends of the latusrectum of the parabola $(y+3)^2 = 8(x-2)$ is
1) (0, -4) 2) (0, -3) 3) (-1, -3) 4) (-2, -3)
48. The point of intersection of the tangents at the points on the parabola $y^2 = 4x$ whose ordinates are 4 and 6 is
1) (6, 5) 2) (7, 3) 3) (9, 10) 4) (6, 10)
49. Equation of the tangent to $x^2 - 4x - 8y + 12 = 0$ at $\left(4, \frac{3}{2}\right)$ is
1) $x + 2y - 1 = 0$ 2) $x + 2y + 1 = 0$
3) $x - 2y + 1 = 0$ 4) $x - 2y - 1 = 0$
50. If the line $y = 3x + 1$ touches the parabola $y^2 = 4ax$ then its length of latusrectum is
1) 24 2) 16 3) 12 4) 18
51. If the tangents to the parabola $y^2 = 4ax$ make complementary angles with the axis of the parabola then $t_1 t_2 =$
1) 0 2) 1 3) -1 4) -2
52. Product of the slopes of the two tangents drawn from (2,3) to $y^2 = 4x$ is
1) $\frac{5}{2}$ 2) $\frac{3}{2}$ 3) 1 4) $\frac{1}{2}$
53. If the tangents at t_1 and t_2 to a parabola are perpendicular then $(t_1 + t_2)^2 - (t_1 - t_2)^2 =$
1) -1 2) -4 3) 6 4) 2
- NORMAL AND ITS APPLICATIONS**
54. If P is a point on the parabola $y^2 = 4ax$ in which the abscissa is equal to ordinate then the equation of the normal at P is
1) $2x + y + 12a = 0$ 2) $2x + y - 12a = 0$
3) $2x + y - 18a = 0$ 4) $x + 2y - 12a = 0$
55. If the line $x - y + k = 0$ is a normal to $y^2 = 4ax$ then the value of k is
1) 4a 2) -a 3) -5a 4) -3a
56. If a normal is drawn to $y^2 = 12x$ making an angle 45° with the axis then the foot of the normal is
1) (3, 8) 2) (3, -6) 3) (12, -12) 4) (8, -8)
57. P(-3, 2) is one end of focal chord PQ of the parabola $y^2 + 4x + 4y = 0$. Then the slope of the normal at Q is
1) -1/2 2) 2 3) 1/2 4) -2
58. Equation of the normal to $y^2 = 4x$ which is perpendicular to $x + 3y + 1 = 0$ is
1) $3x - y - 33 = 0$ 2) $3x - y + 17 = 0$
3) $3x - y + 19 = 0$ 4) $3x - y + 27 = 0$
59. The normal at $(a, 2a)$ on $y^2 = 4ax$ meets the curve again at $(at^2, 2at)$. Then the value of t is equal to
1) 1 2) 3 3) -1 4) -3
60. If the normal chord at 't' on $y^2 = 4ax$ subtends a right angle at vertex then $t^2 =$
1) 0 2) 1 3) 2 4) 3
- CHORD, CHORD OF CONTACT, CHORD LENGTH, MID POINT OF A CHORD**
61. If the length of chord of circle $x^2 + y^2 = 4$ and $y^2 = 4(x - h)$ is maximum, then the value of h is
1) 1 2) -1 3) 4 4) 0
62. If the chord joining the points t_1 and t_2 on the parabola $y^2 = 4ax$ subtends a right angle at its vertex then $t_2 =$
1) $\frac{2}{t_1}$ 2) $\frac{-2}{t_1}$ 3) $\frac{4}{t_1}$ 4) $\frac{-4}{t_1}$
63. The condition for the line $4x + 3y + k = 0$ to intersect $y^2 = 8x$ is
1) $K < \frac{9}{2}$ 2) $K > 5$ 3) $K > \frac{15}{2}$ 4) $K > 6$
64. If the line $5x - 4y - 12 = 0$ meets the parabola $x^2 = 8y$ in A and B then the point of intersection of the two tangents at A and B is
1) (5, 3) 2) (3, 5) 3) (4, 6) 4) (6, 4)
65. A tangent to the parabola $y^2 = 4ax$ meets the axes at A and B. Then the locus of mid point of $\overline{AB}$ is
1) $y^2 + 2ax = 0$ 2) $y^2 - 2ax = 0$
3) $2y^2 + ax = 0$ 4) $4y^2 + ax = 0$

66. Let O be the origin and A be a point on the curve $y^2 = 4x$. Then locus of mid point of OA is

- 1) $x^2 = 4y$ 2) $x^2 = 2y$
3) $y^2 = 16x$ 4) $y^2 = 2x$

67. Length of the chord of contact of (2,5) with respect to $y^2 = 8x$ is

- 1) $\frac{3\sqrt{41}}{2}$ 2) $\frac{2\sqrt{17}}{3}$ 3) $\frac{7\sqrt{3}}{5}$ 4) $\frac{5\sqrt{3}}{7}$

68. If a chord $4y = 3x - 48$ subtends an angle θ at the vertex of the parabola $y^2 = 64x$ then $\tan \theta =$

- 1) $\frac{10}{9}$ 2) $\frac{13}{9}$ 3) $\frac{20}{9}$ 4) $\frac{16}{9}$

ADDITIONAL PROBLEMS

69. Length of the perpendicular dropped from the focus of the parabola $y^2 = -16x$ to a line making the equal intercepts 2 on the positive axes

- 1) $3\sqrt{2}$ 2) $8\sqrt{2}$ 3) $6\sqrt{2}$ 4) $2\sqrt{2}$

70. The ordinate of a point on the parabola $y^2 = 18x$ is one third of its length of the latusrectum. Then the length of subtangent at the point is

- 1) 12 2) 8 3) 6 4) 4

71. If the join of ends of the latusrectum of $x^2 = 8y$ subtends an angle θ at the vertex of the parabola then $\cos \theta =$

- 1) $\frac{-4}{5}$ 2) $\frac{-2}{3}$ 3) $\frac{-3}{5}$ 4) $\frac{-1}{5}$

72. If an equilateral triangle is inscribed in a parabola $y^2 = 12x$ with one of the vertices being at the vertex of the parabola then its height is

- 1) $24\sqrt{3}$ 2) $16\sqrt{3}$ 3) 36 4) 24

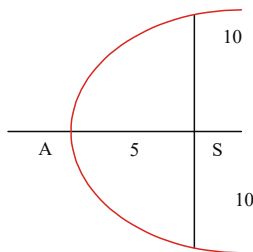
73. If the locus of the point $(4t^2 - 1, 8t - 2)$ represents a parabola then the equation of latusrectum is

- 1) $x - 5 = 0$ 2) $2x - 7 = 0$
3) $x + 5 = 0$ 4) $x - 3 = 0$

LEVEL-I (C.W.) - KEY

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 1) 1 | 2) 4 | 3) 4 | 4) 1 | 5) 3 | 6) 4 |
| 7) 3 | 8) 3 | 9) 1 | 10) 2 | 11) 4 | 12) 2 |
| 13) 1 | 14) 2 | 15) 4 | 16) 4 | 17) 3 | 18) 4 |
| 19) 2 | 20) 2 | 21) 3 | 22) 3 | 23) 1 | 24) 1 |
| 25) 3 | 26) 2 | 27) 3 | 28) 3 | 29) 1 | 30) 1 |
| 31) 4 | 32) 1 | 33) 2 | 34) 2 | 35) 4 | 36) 1 |
| 37) 3 | 38) 4 | 39) 1 | 40) 2 | 41) 2 | 42) 4 |
| 43) 2 | 44) 2 | 45) 3 | 46) 2 | 47) 2 | 48) 1 |
| 49) 4 | 50) 3 | 51) 2 | 52) 4 | 53) 2 | 54) 2 |
| 55) 4 | 56) 2 | 57) 1 | 58) 1 | 59) 4 | 60) 3 |
| 61) 2 | 62) 4 | 63) 1 | 64) 1 | 65) 3 | 66) 4 |
| 67) 1 | 68) 3 | 69) 1 | 70) 4 | 71) 3 | 72) 3 |
| 73) 4 | | | | | |

LEVEL-I (C.W.) - HINTS



1.

Parabola is $y^2 = 4ax \Rightarrow 100 = 4a(5) \Rightarrow a = 5$

Focus lies at a distance of 5 units from vertex.

- $s = (4, 5), A = (3, 6)$ Find $\overline{AS}$
- Distance from S to directrix = $2a$
- Find latus rectum and tangent at vertex is midway line between directrix and latusrectum.
- Find foot of perpendicular Z of S on directrix then vertex is M.P of $\overline{SZ}$
- Focal distance = $x_1 + a$
- Put $(x, 4a)$ on the given curve $y^2 = 4ax$
 $\Rightarrow x = 4a$ and $y = 4a$
 $\frac{\tan \theta}{2} = \frac{y}{x} = \frac{4a}{4a} = 1 \Rightarrow \frac{\theta}{2} = 45^\circ \Rightarrow \theta = 90^\circ$
- $L.L.R = \left| \frac{\text{co. eff } x}{\text{co. eff } y^2} \right| = \frac{4}{2} = 2$
- Partially diff w.r.to 'y'
- Convert into $(y - k)^2 = 4a(x - h)$
- Convert into $(x - h)^2 = 4a(y - k)$

12. Convert into $(y-k)^2 = 4a(x-h)$
13. Focus is $(6, 3)$; L.L.R $= 4a = 8 \Rightarrow a = 2$
vertex is 'a' units from focus along the axis
 $\Rightarrow A = (8, 3)$ or $(4, 3)$
14. $(at^2, 2at) = (4a, 4a) \Rightarrow t = 2$, other end
 $\left(\frac{a}{t^2}, \frac{-2a}{t}\right)$
15. $2at_1 = 4 \Rightarrow t_1 = \frac{1}{2}$;
other end $\left(\frac{a}{t_1^2}, \frac{-2a}{t_1}\right) = (16, -16)$
Now length of focal chord $= \sqrt{225 + 400} = 25$
16. $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$; $S(4, 0) P(1, 4) Q(16, 6)$
 $\frac{SP}{SQ} = \frac{5}{20} = \frac{1}{4}$
17. Partially diff w.r.t 'x'
18. $S(1, 0)$, $m = 1$
Equation $y - 0 = 1(x - 1) \Rightarrow x - y - 1 = 0$
19. The line is directrix of parabola given equation
 $\left(y - \frac{1}{2}\right)^2 = 4\left(x + \frac{17}{16}\right)$ circle will touch the line
 $x = -1 - \frac{17}{16} \Rightarrow 16x + 33 = 0$ (equation of directrix.)
20. The coordinates of focus $S = (a_1, a)$ vertex
 $S = (a_1, a) A = (a, 0); \Rightarrow AS = a_1 - a$
length of Latus rectum $4AS = 4(a_1 - a)$
equation of parabola $y^2 = 4(a_1 - a)(x - a)$
21. $\frac{SL}{SZ} = e \Rightarrow \frac{5/2}{1} = e > 1$
22. write in the form $\Delta \neq 0, h^2 = ab$ and
 $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$
23. Verify the given point in the options when the parabola is of the form $y^2 = 4ax$
24. $SP^2 = PM^2$
25. Y-coordinates are same, Axis is parallel to x-axis
26. $x = ay^2 + by + c$ (Axis is horizontal line)
27. It is of the form $(x - \alpha)^2 = 4a(y - \beta)$ when
 $(\alpha, \beta) = (-3, 4)$, $\therefore 4a = 12$ i.e.,
 $(x + 3)^2 = 12(y - 4)$
29. $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$
 $(2, 0) \in S = 0 \Rightarrow 4 + 4g + c = 0$ & $f^2 = c$
 $\Rightarrow 4 + 4g + f^2 = 0$ $c(-g, -f) = (x, y)$
30. equation of the focal chord having slope 'm' is
 $y - 0 = m(x - 4) \Rightarrow mx - y - 4m = 0$
 $\Rightarrow$ It is tangent to circle $(x - 6)^2 + y^2 = 2$
 $\Rightarrow \left| \frac{6m - 0 - 4m}{\sqrt{m^2 + 1}} \right| = \sqrt{2} \Rightarrow m^2 = 1 \Rightarrow m = \pm 1$
31. $y = x^2 + 6$ diff w.r.t x $\Rightarrow \frac{dy}{dx} = 2x$ at $P(1, 7)$ is 2.
eq of tangent at $P(1, 7)$ to curve is
 $y - 7 = 2(x - 1) \Rightarrow 2x - y + 5 = 0$
Apply the foot of perpendicular of center $(-8, -6)$
on the tangent line $\Rightarrow Q = (-6, -7)$
32. $L(a, 2a)$ and tangent at L is $S_1 = 0$
33. Tangent $y = mx + \frac{a}{m} \Rightarrow y = \frac{1}{\sqrt{3}}x + 2\sqrt{3}$
34. Find tangent $x + y = k$ and area $\frac{k^2}{2}$
35. Convert the line $y = mx + c$, Then tangent condition $c = -am^2$
36. Tangent $y = m(x + a) + \frac{a}{m}$
37. Equation of tangent $yt - x - at^2 = 0$
38. $S_{11} > 0 \Rightarrow$ two tangents
39. Verify $m_1 + m_2 = \frac{y_1}{x_1}$ and $m_1 m_2 = \frac{a}{x_1}$
40. Here $a = 1$ equation of tangent $y = mx + \frac{1}{m}$
it passes through $(-2, -1) \Rightarrow -1 = m(-2) + \frac{1}{m}$

$$\Rightarrow -2m^2 + 1 = -m$$

$$\Rightarrow 2m^2 - m - 1 = 0 \Rightarrow m = 1; -\frac{1}{2}$$

$$\tan \alpha = \frac{1 + \frac{1}{2}}{1 + 1\left(-\frac{1}{2}\right)} = 3$$

41. Apply the formula $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$ = point of contact

43. $(at_1^2, 2at_1) = (x_1, y_1)$ $(at_2^2, 2at_2) = (x_2, y_2)$;

P.I. of tangents $(at_1t_2, a(t_1 + t_2))$ meet on axis

$$t_1 + t_2 = 0 \Rightarrow y_1 + y_2 = 0$$

44. $\sqrt{3}, 1, \frac{1}{\sqrt{3}}$ they are in G.P

45. $\frac{1}{m_1} + \frac{1}{m_2} = c \Rightarrow m_1 + m_2 = cm_1m_2$

46. $y_1 = 2at_1, y_2 = 2at_2, y_3 = a(t_1 + t_2)$

47. They meet on directrix

48. Use $(at_1t_2, a(t_1 + t_2))$

49. Slope $m = \left(\frac{dy}{dx}\right)_{\left(4, \frac{3}{2}\right)}$ or $S_1 = 0$

50. $ln = am^2$

51. Complementary $\Rightarrow m_1m_2 = 1 \Rightarrow t_1t_2 = 1$

52. $m_1m_2 = \frac{a}{x_1}$

53. $t_1t_2 = -1$

54. $at_1^2 = 2at_1 \Rightarrow t_1 = 2$, Normal $y + xt_1 = 2at_1 + at_1^3$

55. Normal condition $al^3 + 2alm^2 + m^2n = 0$

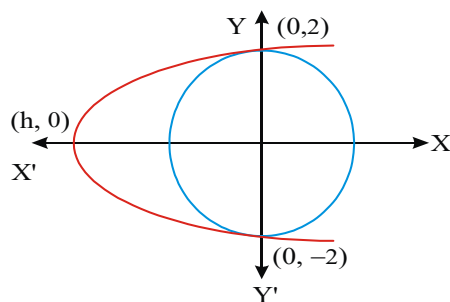
56. Foot of normal $(am^2, -2am)$

57. Slope of tangent at P = Slope of normal at Q

58. Normal condition $al^3 + 2alm^2 + m^2n = 0$

59. $t_2 = -t_1 - \frac{2}{t_1}$

60. $t t_1 = -4$ & $t = -t_1 - \frac{2}{t_1}$



61.

Length of the chord occurs when parabola passing through (0,2) and (0,-2)

$$4 = 4(0 - h) \Rightarrow h = -1$$

62. $PA \perp PB \Rightarrow \frac{2at_1}{at_1^2} \cdot \frac{2at_2}{at_2^2} = -1; \Rightarrow t_1t_2 = -4$

63. $a - mc > 0$ where $y = mx + c$

64. $\left(\frac{-2al}{m}, \frac{n}{m}\right)$

65. Midpoint of $\overline{AB}$ is $P(x_1, y_1)$, Then equation of

$$\overline{AB} \text{ is } \frac{x}{x_1} + \frac{y}{y_1} = 2. \text{ It is tangent to } y^2 = 4ax$$

66. Point A(2x, 2y) lies on $y^2 = 4x$

$$\Rightarrow (2y)^2 = 4(2x) \Rightarrow y^2 = 2x$$

67. Use $\frac{1}{a} \sqrt{(y_1^2 + 4a^2)(y_1^2 - 4ax_1)}$

68. Homogenisation

69. Line $\frac{x}{2} + \frac{y}{2} = 1$ and $S(-4, 0)$

70. Condition: $y_1 = \frac{1}{3}(2a)$; $L.S.T = \left|\frac{y_1}{m}\right|$; where

$$m = \left(\frac{dy}{dx}\right)_{(x_1, y_1)}$$

71. Find the equation of $\overline{OL}$ and $\overline{OL}^\perp$, then find $\cos \theta$

72. $\tan 30^\circ = \frac{y}{x} \Rightarrow x = y\sqrt{3}$ put in the $y^2 = 12x$

$$\Rightarrow x = 36$$

73. $\left. \begin{array}{l} 4t^2 - 1 = x \\ 8t - 2 = y \end{array} \right\} \text{ Eliminate } t$



LEVEL-I (H.W)

**FOCUS, VERTEX, DIRECTRIX,
LATUSECTUM, AXIS, FOCAL CHORD,
FOCAL DISTANCE, DOUBLE ORDINATE**

- The parabola $x^2 = py$ passes through (12,16). Then the focal distance of the point is
 1) $\frac{57}{4}$ 2) $\frac{73}{4}$
 3) 13 4) 18
- If (9,12) is one end of a double ordinate of the parabola $y^2 = 16x$ then its equation is
 1) $x+9=0$ 2) $y+9=0$
 3) $y-9=0$ 4) $x-9=0$
- If $2x+y+a=0$ is a focal chord of the parabola $y^2 = -8x$ then $a =$
 1) 2 2) -2
 3) 4 4) -4
- The focus and directrix of a parabola are (0,0) and $y = 2x+1$. The equation of its axis is
 1) $2x-y-0$ 2) $x+2y=0$
 3) $x+2y=5$ 4) $x+2y-7=0$
- If the vertex and focus of a parabola are (3,6) and (4,5) then the equation of its directrix is
 1) $x-y+7=0$ 2) $x-y+9=0$
 3) $x-y+5=0$ 4) $x-y+3=0$
- Equation of the latusrectum of the parabola $x^2 + 8x + 12y + 4 = 0$ is
 1) $y+3=0$ 2) $y+2=0$
 3) $y+1=0$ 4) $y+5=0$
- One end of latusrectum of the parabola $(x+2)^2 = -4(y+3)$ is
 1) (-4,-6) 2) (-4,6)
 3) (-4,-4) 4) (2,-6)
- A parabola with vertex (2,3) and axis parallel to the y axis passes through (4,5). Then its length of latusrectum is
 1) 5 2) 8 3) 2 4) 6

- The length of the latus rectum of the parabola whose focus is $\left(\frac{u^2}{2g} \sin 2\alpha, \frac{-u^2}{2g} \cos 2\alpha\right)$ and directrix is $y = u^2 / 2g$ is
 1) $\frac{u^2}{g} \cos^2 \alpha$ 2) $\frac{u^2}{g} \cos 2\alpha$
 3) $\frac{2u^2}{g} \cos 2\alpha$ 4) $\frac{2u^2}{g} \cos^2 \alpha$

- The equation of directrix and latusrectum of a parabola are $3x-4y+27=0$ and $3x-4y+2=0$. Then the length of latusrectum is
 1) 5 2) 10 3) 15 4) 20
- Reflection of $y^2 = x$ about y axis is
 1) $x+y^2=0$ 2) $x^2-y=0$
 3) $y^2-4x=0$ 4) $x^2+y=0$
- If $(x_1, y_1), (x_2, y_2)$ are the extremities of a focal chord of the parabola $y^2 = 16x$ then $4x_1x_2 + y_1y_2 =$
 1) -48 2) 0 3) -64 4) 16
- If S is the focus and PQ is a focal chord of the parabola $y^2 = 4ax$ then SP, the semilatusrectum, and SQ are in
 1) AP 2) GP 3) HP 4) AGP
- One end of latusrectum of the parabola $(x+2)^2 = -4(y+3)$ is
 1) (-4,-6) 2) (-4,6) 3) (-4,-4) 4) (2,-6)
- If (4,8) is one end of a focal chord of the parabola $y^2 = 16x$ then its equation is
 1) $y=8$ 2) $x=4$
 3) $x+y=12$ 4) $x-y=16$
- The line $x-1=0$ is the directrix of the parabola $y^2 - kx + 8 = 0$, then one of the values of k is
 1) 1/8 2) 8 3) 4 4) 1/4

EQUATION OF PARABOLA/CONIC

- Equation of the parabola whose vertex is the origin, axis along the y-axis and which passes through (4,2) is
 1) $x^2 = 8y$ 2) $x^2 + 8y = 0$
 3) $y^2 = x$ 4) $y^2 + x = 0$

18. Equation of the parabola having focus (3,-2) and vertex (3,1) is

- 1) $(x-3)^2 = 12(y-1)$ 2) $(x-3)^2 = -12(y-1)$
3) $(y-1)^2 = 12(x-3)$ 4) $(y-1)^2 = -12(x-3)$

19. Equation of the parabola whose axis is vertical and passing through the points (4,5), (-2,11), (-4,21) is

- 1) $x^2 + 6x - 7y + 10 = 0$
2) $x^2 - 6x + 9y + 13 = 0$
3) $x^2 - 4x + 6y - 12 = 0$
4) $x^2 - 4x - 2y + 10 = 0$

20. Equation of a parabola whose vertex is (2,-3), axis is parallel to the x axis and latusrectum 8 is

- 1) $(y+3)^2 = 8(x+2)$ 2) $(y+3)^2 = 16(x-2)$
3) $(y+3)^2 = 8(x-2)$ 4) $(y+3)^2 = 32(x-2)$

21. The point on the parabola $y^2 = 36x$ whose ordinate is three times its abscissa is

- 1) (4,12) 2) (3,9) 3) (6,18) 4) (1,3)

22. Locus of the point equidistant from (0,-1) and the line $y=1$ is

- 1) a parabola with vertex (0,0)
2) a parabola with focus (0,1)
3) a parabola with directrix $y=-1$
4) a parabola with axis $y=0$

23. The point on the parabola $x^2 = y$ which is nearest to (3,0) is

- 1) (1,-1) 2) (-1,1) 3) (-1,-1) 4) (1,1)

TANGENT AND ITS APPLICATIONS

24. Equation of the tangent to $y^2 = 8x$ which is parallel to $x - y + 3 = 0$

- 1) $x - y + 4 = 0$ 2) $x - y + 5 = 0$
3) $x - y + 2 = 0$ 4) $x - y + 7 = 0$

25. Equation of the tangent to $y^2 = 16x$ which is perpendicular to $2x - y + 5 = 0$ is

- 1) $x + 2y + 16 = 0$ 2) $x + 2y - 17 = 0$
3) $x + 2y - 19 = 0$ 4) $x + 2y - 18 = 0$

26. The line among the following that touches the parabola $y^2 = 4ax$ is

- 1) $x + my - am^2 = 0$ 2) $x - my + am^2 = 0$
3) $x + my - am^2 = 0$ 4) $y + mx + am^2 = 0$

27. The line $4x + 6y + 9 = 0$ touches the parabola $y^2 = 4x$ at the point

- 1) $\left(-3, \frac{9}{4}\right)$ 2) $\left(3, \frac{-9}{4}\right)$ 3) $\left(\frac{9}{4}, -3\right)$ 4) $\left(\frac{-9}{4}, -3\right)$

28. Two tangents to the parabola $(y-1)^2 = 4(x-2)$ are at right angles. Then the locus of their point of intersection is

- 1) $x-3=0$ 2) $x-1=0$
3) $2x-1=0$ 4) $3x-1=0$

29. The line $y = m(x+a) + a/m$ touch the parabola $y^2 = 4a(x+a)$ for 'm'

- 1) is equal to zero
2) is any positive real number
3) is any negative number
4) is any real number, $m \neq 0$

30. If the line $x - 3y + k = 0$ touches the parabola $3y^2 = 4x$ then the value of k is

- 1) 5 2) 7 3) 6 4) 3

31. Sum of the slopes of the two tangents drawn from (3,5) to $y^2 = 8x$ is

- 1) $\frac{7}{3}$ 2) 5 3) $\frac{5}{3}$ 4) $\frac{8}{3}$

32. Two tangents to the parabola $y^2 = 4ax$ make supplementary angles with the x-axis. Then the locus of their point of intersection is

- 1) $x=a$ 2) $x=ak$ 3) $y=0$ 4) $x+y=a$

33. The coordinates of the point of intersection of tangents drawn to $y^2 = 4ax$ at the points where it is cut by the line $x \cos \alpha + y \sin \alpha - p = 0$ is

- 1) $(p \tan \alpha, 2a \sec \alpha)$ 2) $(-p \sec \alpha, -2a \tan \alpha)$
3) $(2a \sec \alpha, p \tan \alpha)$ 4) $(p \sec \alpha, 2a \tan \alpha)$

NORMAL AND ITS APPLICATIONS

34. Equation of the normal at $t=4$ to the parabola $y^2 = 6x$ is

- 1) $4x + y - 108 = 0$ 2) $4x + y + 108 = 0$
3) $x + 4y + 108 = 0$ 4) $x + 4y - 108 = 0$

35. The ordinates of the feet of three normals to the parabola $y^2 = 4ax$ from the point $(6a, 0)$ are

- 1) 0, -3a, 3a 2) 0, -2a, 2a
3) 0, -4a, 4a 4) 0, -5a, 5a

36. If the normal to the parabola $y^2 = 4ax$ at $P(1, 2)$ meets the parabola again in Q then $Q =$
 1) $(-6, 9)$ 2) $(9, -6)$ 3) $(-9, -6)$ 4) $(-6, -9)$

CHORD, CHORD OF CONTACT, CHORD LENGTH, MID POINT OF A CHORD

37. Equation of the chord of the parabola $y^2 = 8x$ which is bisected at the point $(2, -3)$ is
 1) $4x + 3y + 1 = 0$ 2) $3x + 4y + 6 = 0$
 3) $2x - 3y - 13 = 0$ 4) $x - y + 5 = 0$
38. The locus of mid points of chords of the parabola $y^2 = 4ax$ passing through the foot of the directrix is
 1) $y^2 = a(x + a)$ 2) $y^2 = 2a(x + a)$
 3) $y^2 = a(x - a)$ 4) $y^2 = 2a(x - a)$
39. If (a, b) is the mid point of the chord of the parabola $y^2 = 4ax$ passing through the vertex then
 1) $a = 2b$ 2) $2a = b$ 3) $a^2 = 2b$ 4) $2a^2 = b^2$
40. Locus of mid points of chords of the parabola $y^2 = 4ax$ parallel to the line $lx + my + n = 0$ is
 1) $2ly + am = 0$ 2) $ly + 4am = 0$
 3) $ly + 3am = 0$ 4) $ly + 2am = 0$
41. If the line $4x + 3y + 1 = 0$ meets the parabola $y^2 = 8x$ then mid point of the chord is
 1) $(-1, 1)$ 2) $(2, -3)$ 3) $(3, -3)$ 4) $(5, -7)$
42. The line $y = (2x + a)$ will not intersect the parabola $y^2 = 2x$ if
 1) $a < \frac{1}{4}$ 2) $a > \frac{1}{4}$ 3) $a = \frac{1}{4}$ 4) $a = -2$
43. The length of the chord $4y = 3x + 8$ of the parabola $y^2 = 8x$ is
 1) $\frac{320}{7}$ 2) $\frac{320}{9}$ 3) $\frac{80}{9}$ 4) $\frac{640}{7}$
44. $y = x\sqrt{2} - 4a\sqrt{2}$ is a normal chord to $y^2 = 4ax$ then its length is
 1) $8a\sqrt{3}$ 2) $4a\sqrt{3}$ 3) $6a\sqrt{3}$ 4) $2a\sqrt{3}$

ADDITIONAL PROBLEMS

45. Equation of the tangent at the vertex of the parabola $x^2 + 4x + 4y + 16 = 0$ is
 1) $y + 3 = 0$ 2) $y + 4 = 0$ 3) $y + 2 = 0$ 4) $y + 1 = 0$
46. If P is a point on the parabola $y^2 = 4ax$ such that the subtangent and subnormal at P are equal then the coordinates of P are
 1) $(a, 2a)$ or $(a, -2a)$
 2) $(2a, 2a\sqrt{2})$ or $(2a, -2a\sqrt{2})$
 3) $(4a, -4a)$ or $(4a, 4a)$ 4) $(5a, -5a)$ or $(5a, -3a)$
47. If O is the vertex and L, L' are the extremities of the latusrectum of the parabola $y^2 = 4ax$ then the area of the triangle OLL' is
 1) $4a^2$ sq. units 2) $2a^2$ sq. units
 3) a^2 sq. units 4) $8a^2$ sq. units
48. Area of the triangle formed by the vertex, focus and one end of latusrectum of the parabola $(x + 2)^2 = -12(y - 1)$
 1) 18 2) 36 3) 12 4) 9
49. Equation of the circle on the latusrectum of $y^2 = 8x$ as ends of diameter is
 1) $x^2 + y^2 - 4y + 16 = 0$
 2) $x^2 + y^2 - 4x - 12 = 0$
 3) $x^2 + y^2 + 4x + 12 = 0$
 4) $x^2 + y^2 - 6x - 12 = 0$
50. If the chord of contact of tangents from P to the parabola $y^2 = 4ax$ touches the circle $x^2 + y^2 = b^2$ then the equation of the locus of P is
 1) $a^2x^2 = b^2(y^2 + 4a^2)$ 2) $4a^2x^2 = b^2(y^2 + 4a^2)$
 3) $2a^2x^2 = b^2(4y^2 + a^2)$ 4) $a^2x^2 = 4b^2(y^2 + a^2)$
51. The area of the triangle inscribed in the parabola $y^2 = 4x$ with the vertices, whose ordinates are 1, 2, 4 is
 1) $\frac{7}{2}$ sq. units 2) $\frac{5}{2}$ sq. units
 3) $\frac{3}{2}$ sq. units 4) $\frac{3}{4}$ sq. units

LEVEL-I (H.W.) - KEY

1)2	2)4	3)3	4)2	5)3	6)2
7)3	8)3	9)4	10)2	11)1	12)2
13)3	14)3	15)2	16)3	17)1	18)2
19)4	20)3	21)1	22)1	23)4	24)3
25)1	26)2	27)3	28)2	29)4	30)4
31)3	32)3	33)2	34)1	35)3	36)2
37)1	38)2	39)4	40)4	41)2	42)2
43)3	44)3	45)1	46)1	47)2	48)4
49)2	50)2	51)4			

LEVEL-I (H.W.) - HINTS

- Find 'P' by $S_{11}=0$ Now, focal distance $=|y_1 + a|$
- Other end is image of P(9,12) w.r.to axis of the parabola. $\therefore$ Its equation is $x=9$.
- $2x + y + a = 0$ passes through focus.
- Axis is perpendicular to directrix and passing through focus.
- Let $Z(x_1, y_1)$ be the foot of the perpendicular drawn from focus to directrix.
Then A is mid point of $\overline{ZS} \Rightarrow Z(2, 7)$ and directrix is perpendicular to SA.
- Equation of the parabola $(x-h)^2 = 4a(y-k)$;
 $(h, k) = (2, 3)$; $a = \frac{1}{2} \Rightarrow 4a = 2$
- Length of Latus rectum $= 2$ (Distance from focus to directrix) Since, length of Latus rectum of the parabola is twice the distance between the focus and the directrix, such that
$$|LR| = 2 \left\{ \frac{\left| -\frac{u^2}{2g} \cos 2\alpha - \frac{u}{2g} \right|}{\sqrt{l^2 + 0^2}} \right\}$$

 $\Rightarrow |LR| = \frac{2u^2}{2g} |1 + \cos 2\alpha| \therefore |LR| = \frac{2u^2}{g} \cos^2 \alpha$
- $2a = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$
- Reflection is $y^2 = -x$
- $(x_1, y_1) = (at_1^2, 2at_1)$; $(x_2, y_2) = (at_2^2, 2at_2)$
- $\frac{1}{SP} + \frac{1}{SQ} = \frac{2}{SL}$
- Verify the options with equation (or) end points of latusrectum lie on the parabola.
- $S(4,0)$ $P(4,8)$; Equation of $\overline{SP}$ is $x=4$
- Equation can be written as $y^2 = k(x - 8/k)$
 $4a = k \Rightarrow a = k/4$; directrix is $x - 8/k + k/4 = 0$
i.e., $x - (8/k - k/4) = 0$ Given equation is $x-1=0$
 $\frac{8}{k} - \frac{k}{4} - 1 = 0 \Rightarrow (k+8)(k-4) = 0 \Rightarrow k = 4, -8$

- X-Coordinates are same., Axis is parallel to y-axis
- $y = ax^2 + bx + c$ (Axis is vertical line)
- Given $y_1 = 3x_1$
- Tangent is $x - y + k = 0$ and condition $In = am^2$
- $x + 2y + k = 0$, Apply tangent condition
- $In = am^2$ for the line $lx + my + n = 0$
- $In = am^2$
- They meet on directrix
- If origin is shifted to (a,0) Then $y^2 = 4ax$ becomes
 $y^2 = 4a(x+a)$ and $y = mx + a/m$
- Tangent condition $In = am^2$
- $m_1 + m_2 = \frac{y_1}{x_1}$
- $\theta_1 = 180 - \theta_2 \Rightarrow m_1 + m_2 = 0$
- $\left(\frac{n}{l}, \frac{-2am}{l} \right)$
- $y + xt = 2at + at^3$
- Normals $at^3 + (2a-x)t - y = 0$
- $t_2 = -t_1 - \frac{2}{t_1}$
- Chord $S_1 = S_{11}$
- $S = S_{11}$ passing through $Z(-a, 0)$
- $S_1 = S_{11}$ passing through vertex (0,0)
- $S_1 = S_{11}$ is parallel to $lx + my + n = 0$. Equating slopes.
- Use $S_1 = S_{11}$
- Apply $cm > a \Rightarrow (a)(2) > \frac{1}{2} \Rightarrow a > \frac{1}{4}$
- Use $\frac{4}{l^2} \sqrt{a(am^2 - \ln)(l^2 + m^2)}$ where the line is
 $lx + my + n = 0$
- Use $\frac{4}{m^2} \sqrt{a(a-mc)(1+m^2)}$
- Convert in $(x-h)^2 = 4a(y-k)$
- Area $= \frac{1}{2} (LL')(AS) = \frac{1}{2} (4a)(a) = 2a^2$
- Area $= \frac{1}{2} (AS)(SL)$
- Centre $S(2,0)$ radius $r = 2a$
- $yy_1 - 2ax - 2ax_1 = 0$ is tangent to $x^2 + y^2 = b^2$
use $d = r$
- Use $A = \frac{1}{8a} |(y_1 - y_2)(y_2 - y_3)(y_3 - y_1)|$